Introduction to lists

A list is a collection of multiple values stored in a single variable
Lists are
* ordered
* mutable (changable)
* can store different datatypes

eg:

```
marks = [85, 90, 78, 92]
data = [10, "python", 3.14, True]
```

Note:
* Each element in a list has a position called an index
* list are mutable, so we can change values
* apped () ⇒ used to add element in list
* remove () ⇒ removes element from list
* len () → returns no of elements in a list

Assignment:
1. Create a list of 5 numbers and print all elements using a loop
2. Take 3 numbers from user and store then in a list. Print the list

List Indexing & Slicing

→ List Indexing:

* Each item has an index (position), starting from '0'

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Negative Indexing:

* Negative index start from end

→ List Slicing:

* Used to extract a portion of the list

syntax:

list [start : end]

* End index is not included

* step in slicing

* step tells how many elements to skip each time.

syntax:

list [start : end : step]

* Reverse using slicing

print (list [: : -1])

problems

1. Pringt last element using negative indexing
2. Slice first 4 elements
3. Revers a list using slicing
4. Iterate the list and print list elements

LIST COMPRESSION

→ A shorter and cleaner way to create list using loops

SYNTAX:
[expression for item in iterable]

with condition:
→ [expression for item in iterable if condition]
→ [expression if condition else for item in iterable]

Function in List:
[func for item in iterable]

Nested List:
[expression for item1 in iterable1 for item2 in iterable2]

Python list Methods

→ pythodn methods helps to do operations on list

1. append()
→ adds element at the end.

2. insert()
→ adds element at a specific index

3. remove()
→ removes first occurance of value

4 pop()
→ removes element by index (defalt last)

5. clear()
→ removes all elements

6. sort()
→ sort list in ascending order

7. reverse()
→ reverse list

8. count()
→counts occurances

9. index()
→returns index of element

Assignment:
[7, 63, 1, 129, 17, 6, 2, 10]
1. Add element using apped()
2. remove element
3. sort a list
4. Clear the List

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List Practic problems

1. Write a program to sum all the elements in a list
2. Write a program to find the largest number in a list
3. Write a program to count how many even numbers are in list
4. Write a program to reverse list
5. Write a program to check if element exists in list

List Compression:

1. Create a list of 1 to 20 that are divisible by 3
2. Replace each number with "Even" or "Odd"
3. Replace all vowels from a string using list comprehension

   "PYTHON PROGRAMMING"